

Six attempts at quasi-polynomial unknot recognition: synthesis, cross-validation, and a faster exact recognizer

Synthesis report prepared with Claude Code

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Abstract

Six independently produced archives each tried to implement the $n^{O(\log n)}$ unknot recognition algorithm announced in Marc Lackenby’s February 2021 talk. This report extracts them into numbered directories, reads their reports, reviews their code, re-runs their test suites, and cross-validates them against each other on shared inputs. All six reach the same honest conclusion: the quasi-polynomial algorithm was not implemented, because the accelerated hierarchy operations named on the last slide have no published algorithmic specification with cost bounds. What they deliver instead is consistent and correct as far as we can test: five reduced-Khovanov recognizers agree on 44 shared inputs, six ball-pattern testers agree on 2000 random spherical patterns, and the grid-diagram search of archive 01 agrees with Khovanov homology on every rectangular diagram we checked. Building on their ideas we add a seventh recognizer, **fastunknot**, whose exact part is a scanning computation of Khovanov homology (Bar-Natan’s local algorithm with delooping and Gaussian elimination) preceded by an Alexander-polynomial filter. It is still exponential in the worst case, but its cost is governed by the width of a scan through the diagram rather than by the crossing number: the 80-crossing unknot family on which every cube implementation needs 2^{80} states is decided in eleven milliseconds. The $n^{O(\log n)}$ target remains open; the final sections state precisely what is missing.

Contents

1	Task, sources, and outcome	2
1.1	What was asked	2
1.2	The talk	2
1.3	The 2026 preprint	3
1.4	Outcome in one paragraph	3
2	The six archives	3
2.1	Extraction	3
2.2	What each archive delivers	4
2.3	Common structure of the six reports	5
3	Code review	5
3.1	The Khovanov recognizers (02–06)	5
3.2	The pattern testers (all six)	6
3.3	Normal coordinates and potentials	6
3.4	The grid search of archive 01	6

4	Cross-validation	6
4.1	Five Khovanov implementations on shared inputs	6
4.2	Six pattern testers on random spherical patterns	7
4.3	Grid search versus Khovanov homology	7
4.4	What the cross-validation does and does not show	8
5	What is missing for $n^{O(\log n)}$	8
6	A faster exact recognizer: <code>fastunknot</code>	9
6.1	Design	9
6.2	The scanning computation	9
6.3	Validation	10
6.4	Measurements	10
6.5	Limits of the new package	11
7	Assessment and recommendations	12
A	Files produced	12

1 Task, sources, and outcome

1.1 What was asked

The original request to each of the six runs was an implementation of the main theorem of the supplied talk [1]: an algorithm that decides whether an n -crossing knot diagram represents the unknot in time $n^{O(\log n)}$. The present task was to unpack the six resulting archives, read their reports, review their code, and synthesize what was achieved and what obstacles remain. A second task was added mid-way: write the asymptotically fastest algorithm possible, trying to reach $n^{O(\log n)}$.

1.2 The talk

The supplied file `docs/quasipolynomial-talk.pdf` has 109 pages including progressive-reveal duplicates. Its logical content is short:

- pp. 11–14: the theorem, $n^{O(\log n)}$ for an n -crossing diagram;
- pp. 15–30: hierarchies, boundary patterns, essential patterns (a proper disc meeting the pattern at most three times must cut off an empty disc, an arc, or a tripod), and the Waldhausen–Johannson characterization: the knot is nontrivial iff the exterior admits an essential hierarchy ending in balls;
- pp. 38–53: simplification of an inessential hierarchy by a violating disc (compress or pattern-compress the last surface the disc runs over, discard later surfaces), and the basic algorithm as a flowchart;
- pp. 54–65: termination by a lexicographic complexity $(g(S_1), \dots, g(S_\ell))$, with the warning that “genus” may mean χ_- or a pattern-sensitive variant, and the estimate “running time at most Lg^L ” if every surface has complexity at most g and hierarchies have length at most L ;
- pp. 66–74: the four speed-ups needed for $n^{O(\log n)}$: (1) surfaces with $g \leq O(n^2)$, (2) compressed hierarchy encodings (Agol–Hass–Thurston), (3) multi-surfaces, (4) generalised Heegaard splittings and Cheeger regions, “the effect of this is to ensure that $L \leq O(\log n)$ ”;

- pp. 75–108: sketches of each speed-up, including the Cheeger-region condition $g(\partial M_i) \leq \frac{1}{3} \min\{g(H_i), g(H) - g(H_i)\}$;
- p. 109: the final flowchart with the boxes BUILD HIERARCHICAL MULTI-SURFACE, CUT ALONG SURFACE, USE CHEEGER REGION, HAKEN’S LEMMA, WEAKLY REDUCE, SIMPLIFY MULTI-SURFACE, and the terminal test “does the graph dual to the boundary pattern have a violating cycle?”.

The slides describe the strategy; they contain no data structures, no algorithms for the boxes, and no proof of the depth bound.

1.3 The 2026 preprint

All six archives located Lackenby’s July 2026 preprint [2] (“Incompressible surfaces, hierarchies and unknot recognition”, 54 pages). We downloaded it and read Section 9 (“The number of steps”) ourselves. It opens: “There are various steps in the algorithm from Theorem 1.3 that are not described with enough precision to be able to estimate their running time. So, we will initially content ourselves with an analysis of the number of times that the algorithm goes through any of the steps”, in terms of the maximal hierarchy length L and the maximal pattern-complexity g , and “At present, we will not attempt to bound these two quantities.” Proposition 9.1 then gives the iteration bound $L(g+1)^L$ using exactly the base- $(g+1)$ digit argument that the talk sketches in base g . Section 10 refines the algorithm to obtain a bound on L that is *linear* in the initial q -complexity of the handle structure, not logarithmic in n . The preprint also supplies two facts the archives use: Proposition 2.6 (a pattern on a 3-ball is essential iff it is empty, a circle, or a connected graph whose planar dual is simple and 4-connected or K_3 or K_4) and Definition 3.4 (pattern-complexity $\chi_P(S) = -4\chi(S) + |S \cap P|$).

So the accelerated, logarithmic-depth algorithm of the talk is, as of September 2026, announced but not specified in any source available to us. This is the single obstacle behind every “partial” verdict below, and it is mathematical, not an engineering shortfall.

1.4 Outcome in one paragraph

None of the six archives, and not the new package either, implements a recognizer with a proved $n^{O(\log n)}$ bound. Each archive contains a complete exact recognizer with an exponential bound, plus genuinely implemented polynomial pieces of the hierarchy strategy (the terminal ball-pattern test, cocycle-dual normal coordinates, the potential arithmetic), and each report says so without hedging. We verified every test suite, cross-validated the archives against each other and against independent computations, found no incorrect verdict, and identified one gap in a correctness argument (Section 3.4). The new package **fastunknot** improves the exact part substantially in practice (Section 6); its worst case is still $2^{O(n)}$.

2 The six archives

2.1 Extraction

The six ZIP files in **reports/** were extracted into **reports/01** through **reports/06**, each with the single wrapper directory removed, in the order of the archive modification times:

Dir	archive	package	Python lines	tests
01	unknot_recognition_implementation.zip	unknot (grid search)	1753	40
02	unknot_recognition_partial_implementation.zip	unknotlab	1939	64
03	unknot_workbench.zip	unknot_lab	1431	49
04	unknot_recognition_partial.zip	unknot	1422	62
05	unknot_recognition_partial_implementation (1).zip	unknot_recognition	1882	58
06	unknot-recognition-implementation.zip	unknot	1278	45

Every archive contains a README, a LaTeX report with PDF, examples, recorded results, and a standard-library-only Python package. All six test suites were re-run here on CPython 3.14.4 under Windows (the archives recorded CPython 3.13.5 on Linux) and pass with exactly the claimed number of tests.

2.2 What each archive delivers

Archive 01: rectangular diagrams and Dynnikov moves. The only archive that does not use Khovanov homology. It represents knots as grid diagrams, searches the finite space of grids reachable by non-increasing moves (exchanges and destabilizations, modulo cyclic shifts) and relies on Dynnikov’s monotonic simplification theorem [5]: an unknot reaches the 2×2 grid, a knot is certified by a state set closed under all moves. Certificates (positive path, non-unit determinant, closed set) are replayed by a verifier that enumerates moves independently. Bound $2^{O(g \log g)}$ in the grid size g ; the report is careful that g is not the crossing number. It validated all 44 719 connected grids of size ≤ 6 (1285 shift orbits). It also contains the pattern test, exact rational H^1 of a simplicial complex, cocycle-dual normal coordinates, and the potential arithmetic.

Archive 02: Khovanov cube with fast certificates. Descending-diagram test (a diagram traversed meeting every crossing first on its over-strand is trivial), Fox determinant, then the full reduced Khovanov complex over $\mathbb{F}_2$. Its normal-surface module accepts genuine face-pairing triangulations, checks orientability and sphere/disc vertex links, and computes cocycle-dual normal coordinates without expanding discs. Six braid/determinant pairs were checked against Wolfram KnotData.

Archive 03: workbench with an explicit exponential lower bound. Khovanov cube with default resource ceilings (65 536 states), a bond-enumeration pattern test with checkable witnesses, and a `bound` command for the conditional $L(g+1)^L$ arithmetic. Its report proves that its own implementation is not quasi-polynomial: on the unknot $\sigma_1 \overline{\sigma_2} \cdots \sigma_n$ the cube has 2^n states and 3^n generators, and the determinant filter cannot help since the determinant is 1.

Archive 04: Khovanov cube with Knot Atlas fixtures. Khovanov cube, dual-4-connectivity pattern test, potential arithmetic. It checks $10_{124} = T(3, 5)$ from an Atlas PD code and contains a clean chain-level proof that the unreduced $\mathbb{F}_2$ complex is two copies of the reduced one, which justifies reading reduced ranks off published unreduced tables.

Archive 05: Khovanov cube with Reidemeister preprocessing. Crossing-decreasing Reidemeister I/II moves with replayable traces, determinant filter, Khovanov cube; bond-enumeration pattern test; simplicial normal coordinates with manifold validation; the pattern-complexity $-4\chi + |S \cap P|$ of the preprint. It exhausts all 682 two-strand words of odd length ≤ 9 and gives an independent braid-theoretic proof that its eight-crossing “hard unknot” is trivial.

Archive 06: Khovanov cube with 11-crossing tests. Khovanov cube, bond-enumeration pattern test, a `verify` command that recomputes a report with $d^2 = 0$ checks. Its fixtures include the Kinoshita–Terasaka and Conway knots (Alexander polynomial 1, reduced rank 33). Its source audit notes that a 34-page Oxford version of the slides uses the factor $1/10$ instead of $1/3$ in the Cheeger condition and displays $O((\log n)^2)$ for the hierarchy length; substituting that depth into the digit count would give $n^{O((\log n)^2)}$.

2.3 Common structure of the six reports

Read side by side, the reports agree on every substantive point, which is itself evidence since they were produced independently:

1. The requested bound is not established; the delivered recognizer is a different, exponential algorithm; resource limits return UNKNOWN and never a verdict.
2. The Khovanov route is justified by Kronheimer–Mrowka [3] plus a universal-coefficient argument: $\dim_{\mathbb{F}_2} \widehat{\text{Kh}}(K; \mathbb{F}_2) \geq \text{rank } \widehat{\text{Kh}}(K; \mathbb{Z})$, and the rational reduced homology is nonzero because its Euler characteristic is the reduced Jones polynomial, so $\mathbb{F}_2$ -rank one forces rational rank one.
3. Determinant one is never a positive certificate ($T(3, 5)$ and the Kinoshita–Terasaka knot are the standard witnesses).
4. The talk’s “base g ” digit argument needs base $g + 1$ when the digit g is allowed, since $(1, 0)$ and $(0, g)$ have the same base- g value; all six use base $g + 1$, as does Proposition 9.1 of the preprint.
5. The terminal pattern test is implemented only under an explicit “already a 3-ball” hypothesis, and $b_1 = 0$ is not ball recognition ($S^2 \times I$ is the standard counterexample).
6. The missing operations are listed by name: initial layered handle structure of the exterior, compressed cutting with provenance, hierarchical multi-surfaces with the pattern-sensitive quadratic bound, lifting a violating dual cycle to a disc in an earlier stage, Cheeger-region detection and Heegaard modification, Haken’s lemma and weak reduction, and an end-to-end bit-cost proof including restarts.

3 Code review

3.1 The Khovanov recognizers (02–06)

All five have the same architecture, evidently converged upon rather than copied (variable names, data layouts and gradings differ): enumerate the 2^n resolutions with a disjoint-set structure, encode a reduced generator as a bit mask over the unmarked circles with the marked circle fixed to x , build each differential column as a Python integer (XOR arithmetic over $\mathbb{F}_2$), and compute ranks by pivot elimination. Points checked in the review:

- Input validation is uniformly careful: one component, spherical rotation system via $V - E + F = 2$ on the ribbon graph (so virtual diagrams are rejected, not misclassified), labels normalized, `{"pd": []}` defined as one circle.
- The Frobenius algebra tables ($m(x \otimes x) = 0$, $\Delta(1) = 1 \otimes x + x \otimes 1$, $\Delta(x) = x \otimes x$) are identical everywhere, and every implementation checks that the marked circle keeps its x label.
- $d^2 = 0$ is available as an exhaustive optional check in all five and is exercised by their tests.
- Gradings are deliberately unnormalized; only the total rank is used, so writhe shifts are irrelevant. This is correct for recognition.
- Resource semantics are consistent (UNKNOWN, exit code 2 or 3; the codes differ between archives, which is a minor interoperability wart).
- Archive 03 imposes default ceilings, so its default invocation is incomplete above 16 crossings; the README says so.

We found no correctness problem. The determinant filters (Fox colouring matrix, Bareiss elimina-

tion with exact-division checks) are also equivalent.

3.2 The pattern testers (all six)

Three different criteria are implemented for “is this cubic spherical pattern essential on a ball”: dual simple and 4-connected with the K_3 , K_4 exceptions (01, 04), no dual loop, parallel pair or non-facial dual triangle (02), and no bond of size 1 or 2 and no three-edge bond with more than one vertex on each side (03, 05, 06). Each report proves its own criterion from the definition. The proofs are correct; the bond argument in archive 06 (the cycle rank of a side with k vertices and r cut edges is $(k - r + 2)/2$) is the cleanest. Input conventions differ (darts $2e, 2e + 1$; signed edge ends; positional darts; edge identifiers occurring twice), which is why a converter was needed for cross-validation.

3.3 Normal coordinates and potentials

Archives 01, 02 and 05 compute integral cocycles spanning $H^1(\cdot; \mathbb{Q})$ in a spanning-tree gauge and convert a cocycle to normal coordinates by half-integral level sets of local vertex heights. The constructions agree (seven coordinates per tetrahedron, matching equations checked, Euler characteristic without disc expansion). All are honest that this is not a saturated $\mathbb{Z}$ -basis, not a connected surface, and not a cut. The four potential modules implement the same base- $(g + 1)$ lemma and differ only in interface.

3.4 The grid search of archive 01

This is the one place where the review found a gap in an argument. The recognizer’s negative certificate is a set of grids closed under all implemented moves; soundness needs the implemented move set to be exactly a set for which monotonic simplification holds. The report states that “rows sharing an endpoint are not accepted by this implementation’s exchange rule” and cites Dynnikov’s Proposition 6, but does not argue that Dynnikov’s theorem still holds when exchanges of two adjacent rows (or columns) with a common corner column are excluded. Depending on the reading of Dynnikov’s definition of interleaving, that exclusion may remove legal moves, in which case a closed set could exist for an unknot. We could not settle this from the sources at hand, so we tested it: Section 4.3 compares the search verdict with Khovanov homology on every grid of size ≤ 6 and on random larger grids, and found no disagreement. The gap is in the proof, not (as far as tested) in the answers.

Everything else in archive 01 checked out: the move implementation, the Booth canonicalization, the independent verifier, and the exhaustive counts ($g!(g - 1)!/2$ connected grids per size).

4 Cross-validation

None of the archives compared itself with another; each ran its own tests. We wrote three scripts (in `synthesis/data/`) that load the packages side by side.

4.1 Five Khovanov implementations on shared inputs

The same braid words and Knot Atlas PD codes were passed to all five packages (after converting label conventions). Reduced $\mathbb{F}_2$ ranks:

Input	02	03	04	05	06	agree
unknot circle	1	1	1	1	1	yes
trefoil	3	3	3	3	3	yes
figure-eight	5	5	5	5	5	yes
T(2,5)	5	5	5	5	5	yes
T(2,7)	7	7	7	7	7	yes
T(3,4)	5	5	5	5	5	yes
T(3,5) det 1	7	7	7	7	7	yes
unknot s1 s2	1	1	1	1	1	yes
unknot s1s2s3s4	1	1	1	1	1	yes
conjugated unknot	1	1	1	1	1	yes
hard unknot 05	1	1	1	1	1	yes
KT 11n42 (Atlas PD)	33	33	33	33	33	yes
Conway 11n34 (Atlas PD)	33	33	33	33	33	yes
10_124 (Atlas PD)	7	7	7	7	7	yes
30 random braid closures (3–4 strands, 6–10 letters)	ranks 1–13					30/30

All 44 inputs agree, including the two Alexander-polynomial-one 11-crossing knots and 30 random closures whose ranks range from 1 to 13.

4.2 Six pattern testers on random spherical patterns

Random perfect matchings of the darts of V trivalent vertices were kept when every component’s rotation system is spherical, converted to the six input formats, and classified by all six testers (400 distinct systems per vertex count) together with six named patterns:

Vertices	spherical systems tested	random matchings drawn	essential	disagreements
2	400	883	102	0
4	400	737	12	0
6	400	1179	0	0
8	400	2603	0	0
10	400	6144	0	0

Pattern	verdict	all six agree
theta	essential	yes
tetrahedron	essential	yes
cube	essential	yes
prism	inessential	yes
dodecahedron	essential	yes
dumbbell	inessential	yes

No disagreement among 2006 patterns and three different criteria. (Random matchings on many vertices are almost always inessential because short bonds are common; the named cube and dodecahedron are the essential large examples.)

4.3 Grid search versus Khovanov homology

We wrote an independent grid-to-PD converter (traversal of the grid, crossing signs from the directions of the vertical and horizontal strands) and compared archive 01’s search (determinant filter disabled) with archive 04’s Khovanov rank:

Grid size	sample	diagrams	unknots	knots	disagreements
2	exhaustive (all shift orbits)	1	1	0	0
3	exhaustive (all shift orbits)	2	2	0	0
4	exhaustive (all shift orbits)	6	6	0	0
5	exhaustive (all shift orbits)	64	62	2	0
6	exhaustive (all shift orbits)	1212	1185	27	0
7	random, ≤ 14 crossings	300	286	14	0
8	random, ≤ 14 crossings	200	188	12	0
9	random, ≤ 14 crossings	100	82	18	0

The knotted orbits of size 5 are the two trefoils (grid number 5) and those of size 6 are 26 trefoil presentations and one figure-eight, exactly as the grid-number theory predicts. No disagreement was found, which supports but does not prove the completeness of the restricted move set. The sample stops at size 9 because the search itself becomes expensive there (one size-9 grid needed 900 863 states, and the 100 grids took 1.9 hours), while the Khovanov side stayed below 14 crossings by construction.

4.4 What the cross-validation does and does not show

Agreement of five independently written cubes is strong evidence that all five compute the same invariant, and agreement with the published ranks (trefoil 3, figure-eight 5, $T(3, 5)$ 7, $K11n34$ and $K11n42$ 33) ties that invariant to reduced Khovanov homology. It says nothing about the quasi-polynomial question, and all runs are on diagrams of at most 14 crossings because the cubes cannot go further.

5 What is missing for $n^{O(\log n)}$

Consolidating the six audits with our own reading of the sources, an implementation with a proved $n^{O(\log n)}$ bound would require the following, none of which exists in any archive or, to our knowledge, in print.

1. **A representation.** A layered handle structure of the knot exterior with a Morse function, boundary-pattern labels, the ambient embedding in S^3 (used by the generalised Seifert construction of pp. 75–78), compressed normal-surface data, and regluing maps, with a proved bound on the bit size of all of this after arbitrarily many operations. The archives’ normal-coordinate modules are local ingredients only.
2. **Bounded multi-surfaces.** An algorithm producing independent relative homology classes with embedded representatives whose *pattern-sensitive* complexity is $O(n^2)$, and a proof that compression and pattern-compression preserve the invariants that allow later surfaces to be kept. A rational H^1 basis is not this.
3. **Compressed cutting.** Cutting, component extraction, Euler characteristic, orientation and pattern transport on compressed data without expanding parallel discs, with provenance sufficient to lift a violating dual cycle in a terminal ball to a disc in the correct earlier stage. Agol–Hass–Thurston orbit counting is a tool, not this procedure.
4. **Cheeger regions and weak reduction.** Recognition of the Heegaard–Morse hypotheses, construction of the modified generalised splitting, Haken’s lemma and weak reduction, and a global progress measure bounding restarts. The inequality $3g(\partial M_i) \leq \min\{g(H_i), g(H) - g(H_i)\}$ is a test on three integers, not the operation.
5. **The depth bound.** A proof that the executable process, after every rebuild and restart, has hierarchy length $O(\log n)$ (the talk) or $O((\log n)^2)$ (the Oxford version cited by archive 06), together with polynomial digit bounds. The preprint’s Section 10 gives a linear bound in the initial q -complexity, and its Section 9 explicitly declines to bound g and L .

6. **The cost accounting.** Only after all of the above does the conditional lemma proved in every archive, $L(g+1)^L$ iterations times a polynomial cost per iteration times polynomially many restarts, become a running-time theorem: $\log((g+1)^L) = O((\log n)^2)$.

The archives also list, and we agree with, the substitutions that do not work: an exponential search with a time cap is not a total algorithm; a polynomial iteration count does not bound work per iteration; expanding a compressed normal surface loses the compression; $b_1 = 0$ is not ball recognition; a Jones or Alexander test does not detect the unknot.

6 A faster exact recognizer: **fastunknot**

6.1 Design

Since the geometric route cannot be completed from the available sources, the second task was addressed by making the *exact* route as fast as we could while keeping every verdict provable. The package **fast/fastunknot** (standard library only, Python ≥ 3.10) runs:

1. validation (PD codes, braid closures, or grid diagrams in the archive-01 convention, converted by the traversal of Section 4.3);
2. crossing-decreasing Reidemeister I/II moves, as in archive 05;
3. the descending-diagram test of archive 02;
4. the *Alexander polynomial*, computed exactly by fraction-free Bareiss elimination over $\mathbb{Z}[t]$ on a first minor of the Fox matrix of the Wirtinger presentation (rows $(1-t)x_o + tx_u - x_v$ at positive and $(1-t)x_o - x_u + tx_v$ at negative crossings, with signs from the traversal); $\Delta \neq 1$ certifies knottedness and strictly dominates the determinant filter of the archives ($T(3,5)$ and all torus knots are now rejected in polynomial time; the Kinoshita–Terasaka and Conway knots are not);
5. reduced Khovanov rank over $\mathbb{F}_2$ by *scanning*.

6.2 The scanning computation

This is Bar-Natan’s local algorithm [4] specialised to $\mathbb{F}_2$ and implemented directly in terms of the arc-algebra description of morphisms. The crossings are processed one at a time in a greedy order chosen (from up to twelve starting crossings) to keep the boundary of the processed region small. After k crossings the partial complex has:

- objects: perfect matchings of the boundary edges, each with a homological degree; closed circles are delooped, one object with a circle becoming two objects (labels 1 and x);
- morphisms $a \rightarrow b$: $\mathbb{F}_2$ -combinations of dot-subsets of the circles of $a \cup \bar{b}$, i.e. elements of $A^{\otimes \text{circles}}$ with $A = \mathbb{F}_2[x]/(x^2)$.

Composition is the two-dimensional TQFT of the glued surface: a component of genus ≥ 1 or with two dots is zero (because the handle operator $m \circ \Delta = 2x$ and x^2 vanish), a closed component is 1 exactly when it carries one dot, and a planar component with β boundary circles and $d \in \{0, 1\}$ dots is $\Delta^{(\beta-1)}(x^d)$, which over $\mathbb{F}_2$ is $x \otimes \cdots \otimes x$ for $d = 1$ and the sum over one undotted position for $d = 0$. Delooping is the same computation with a cup on the source circle (dotted for label x) and a cap on the target circle (dotted to extract the coefficient of 1, since the dual basis of $\{1, x\}$ under the counit is $\{\varepsilon(x \cdot), \varepsilon\}$). After every crossing all invertible entries are cancelled: an entry $b \rightarrow c$ between equal matchings is a unit of the local ring $\text{End}(a) \cong A^{\otimes k}$ iff its expansion contains the undotted term, its inverse is the finite geometric series of the nilpotent part, and Bar-Natan’s cancellation lemma replaces ε by $\varepsilon - \gamma\varphi^{-1}\delta$. When the last crossing is added the boundary is empty, every morphism is a scalar, so a minimal complex has zero differential and its number of objects is the unreduced total rank; the reduced rank is half of it (archive 04’s two-copies lemma) and equals 1 exactly for the unknot.

Proposition 6.1 (Correctness). *On a validated one-component diagram, `fastunknot` returns `UNKNOT` iff the knot is trivial, unless a user-selected ceiling stops it, in which case it returns `UNKNOWN`.*

Proof. Steps 2–3 are isotopies and a sufficient triviality criterion; step 4 uses $\Delta_{\text{unknot}} = 1$. Step 5 computes the homology of the Khovanov complex: the tensor product of crossing complexes in Bar-Natan’s category, delooping, and Gaussian elimination are chain homotopy equivalences, and the closed complex is the standard one. The decision rule is the detection theorem with the universal-coefficient argument of the archives. $\square$

Proposition 6.2 (Cost). *Steps 1–4 are polynomial in n . Step 5 is $2^{O(n)}$ in the worst case.*

Proof. The Alexander determinant is $O(n^3)$ polynomial operations of degree $\leq n$ with polynomially bounded coefficients. In step 5 the objects before elimination after crossing k number at most $2 \cdot 2^c$ times the objects after crossing $k - 1$, where $c \leq 2$ closed circles can appear, so the whole computation is dominated by the cube; compositions cost polynomial time in the boundary size. $\square$

The interesting quantity is the size after elimination. Empirically it stays close to (number of crossingless matchings of the boundary) $\times$ (a multiplicity that grows with the tangle’s homology), i.e. roughly $2^{O(w)} \cdot \text{poly}$ for scan width w . **We have no proof of this**, and there cannot be one in general for the multiplicity: the reduced rank of $\#_k K11n34$ is 33^k with Alexander polynomial 1, so any method that outputs the full homology is exponential on such inputs, whatever the scan width. A subexponential unknot recognizer along these lines would have to detect “rank > 1 ” without finishing the computation; we do not know how.

6.3 Validation

The package has 19 unit tests (polynomial arithmetic; Alexander polynomials of the trefoil, figure-eight, 5_2 , $T(3, 5)$, $\Delta = 1$ for the two 11-crossing knots and two hard unknots; symmetry $\Delta(t) = \Delta(t^{-1})$, $|\Delta(1)| = 1$, mirror invariance and odd determinant on 40 random braids; Reidemeister moves; the local endomorphism ring; $d^2 = 0$ after every crossing; independence of the processing order; resource-limit semantics; known ranks; an independent dense reduced-cube computation on 25 random braids; the command line). Beyond the tests, the scanner agrees with archive 04’s cube on 120 random braid closures (3–5 strands, 6–13 letters, with random processing orders), and reproduces the Atlas ranks 33, 33 and 7.

6.4 Measurements

Timings are on the same Windows machine as everything else in this report; they illustrate the dependence on scan width and are not complexity evidence.

Family	n	reduced rank	max. boundary	objects before	objects after	seconds
unknot sigma_1..sigma_n	8	1	2	6	2	0.001
unknot sigma_1..sigma_n	16	1	2	6	2	0.004
unknot sigma_1..sigma_n	32	1	2	6	2	0.013
unknot sigma_1..sigma_n	64	1	2	6	2	0.036
unknot sigma_1..sigma_n	128	1	2	6	2	0.133
unknot sigma_1..sigma_n	256	1	2	6	2	0.475
T(2,n)	5	5	4	30	10	0.003
T(2,n)	11	11	4	66	22	0.011
T(2,n)	21	21	4	126	42	0.043
T(2,n)	41	41	4	246	82	0.153
T(3,n)	8	5	6	42	10	0.010
T(3,n)	10	7	6	54	14	0.022
T(3,n)	14	9	6	78	18	0.032
T(3,n)	16	11	6	90	22	0.045
T(3,n)	20	13	6	114	26	0.092
T(3,n)	22	15	6	126	30	0.089
random 3-braid	20	75	6	762	150	0.217
random 3-braid	40	321	6	3354	642	1.035
random 4-braid	15	5	6	30	10	0.015
random 4-braid	21	25	8	438	146	0.576
random 4-braid	31	45	8	604	216	0.947
random 4-braid	41	109	8	654	218	2.683
random 5-braid	24	13	6	109	42	0.099
random 5-braid	36	timeout	timeout	timeout	timeout	600.147
random 6-braid	31	85	8	548	216	1.194

Example	n	verdict	deciding step	seconds
conway.json	11	KNOTTED	reduced-khovanov-F2-scan	0.0528
figure_eight.json	4	KNOTTED	alexander-polynomial	0.0001
grid_determinant_one_knot.json	10	KNOTTED	alexander-polynomial	0.0010
grid_scrambled_unknot.json	6	UNKNOT	reidemeister-reduction	0.0004
hard_unknot_8.json	8	UNKNOT	reduced-khovanov-F2-scan	0.0022
kinoshita_terasaka.json	11	KNOTTED	reduced-khovanov-F2-scan	0.0593
torus_3_5.json	10	KNOTTED	alexander-polynomial	0.0009
trefoil.json	3	KNOTTED	alexander-polynomial	0.0001
unknot.json	0	UNKNOT	reidemeister-reduction	0.0000
unknot_braid40.json	40	UNKNOT	reidemeister-reduction	0.0090

One case timed out: a random 36-letter closure on five strands exceeded the 600-second budget, while a 31-letter closure on six strands took about a second. Both scans have boundary at most ten edges; the difference is the size of the minimal complexes of the intermediate tangles, the multiplicity factor discussed above, which is exactly the unproved part of the cost.

Two comparisons with the archives' cubes on identical inputs: the Kinoshita–Terasaka knot (11 crossings, 2048 states, 11 559 generators) takes 0.24–0.31 s in the cubes and 0.07 s in the scan with a boundary of six edges and at most 66 objects after elimination; the unknot $\widehat{\sigma_1 \cdots \sigma_n}$, archive 03's explicit exponential family, needs 2^n states in every cube and is decided by the scan in 0.011 s for $n = 80$ with two objects after every elimination.

6.5 Limits of the new package

It is a reference implementation in pure Python; a compiled implementation of the same algorithm (as in Bar-Natan's or the KnotJob programs) would be far faster. Its scan-order heuristic is greedy;

for diagrams whose Tait graph is grid-like the width is $\Theta(\sqrt{n})$ at best and the minimal complexes can be large. It does not produce a spanning disc or a Reidemeister sequence. And, to repeat, it is not quasi-polynomial.

7 Assessment and recommendations

What was achieved. Seven mutually consistent exact recognizers; a verified and cross-validated body of code for the terminal ball-pattern test and for cocycle-dual normal surfaces; a precise, sourced statement of the conditional $L(g+1)^L$ accounting and of everything a $n^{O(\log n)}$ implementation would still need; and one recognizer whose practical reach is well beyond the 2^n -state cubes.

Obstacles that remain. The six items of Section 5. They are research problems in 3-manifold topology and algorithm design, not coding tasks: the specification of the accelerated hierarchy exists only as a sketch on slides, and the one detailed paper on the subject states that its steps are not specified precisely enough for a running-time estimate. Until that changes, any “ $n^{O(\log n)}$ implementation” would be an exponential search with a mislabelled bound, which is exactly what all six archives refused to deliver.

If work continues. (1) Track the published record: a full version of the 2021 announcement, or the refined Section 10 algorithm with an explicit cost model, would change the picture. (2) Settle the move-set question of Section 3.4 from Dynnikov’s paper, or add shared-corner exchanges to archive 01 and re-run its exhaustive validation. (3) Port the scanning recognizer to a compiled language and replace the greedy order by a tree-decomposition-based one; the planar separator theorem gives width $O(\sqrt{n})$ for the projection graph, which would make the empirical behaviour subexponential even if unproved. (4) Use the archives’ exact recognizers as oracles for any future geometric implementation, as several of their reports suggest.

References

- [1] Marc Lackenby, *Unknot recognition in quasi-polynomial time*, slides, February 2021, 109 pages, supplied as docs/quasipolynomial-talk.pdf.
- [2] Marc Lackenby, *Incompressible surfaces, hierarchies and unknot recognition*, arXiv:2607.23350v1, 25 July 2026, 54 pages. <https://arxiv.org/abs/2607.23350>.
- [3] P. B. Kronheimer and T. S. Mrowka, *Khovanov homology is an unknot-detector*, Publ. Math. IHÉS 113 (2011), 97–208.
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A Files produced

- reports/01 – reports/06: the extracted archives (flat, one level).

- `synthesis/report.tex`, `synthesis/report.pdf`: this report; `synthesis/make_tables.py` regenerates the tables from `synthesis/data/*.json`; the cross-validation scripts are in `synthesis/data/`.
- `fast/`: the `fastunknot` package, tests, examples, `benchmark.py` and `results/benchmark.json`.